



University of Sri Jayewardenepura
Faculty of Engineering

Performance Evaluation System (PES)

An Intelligent Academic Management Platform with AI-Powered Insights

Data Management Project Proposal 2026

01 Introduction

A comprehensive web-based platform designed to revolutionize how students interact with their academic journey at the Faculty.

Academic Tracking

Complete course history, results, and CGPA displayed semester-wise, Time tables
Enrollments

Attendance Monitoring

Real-time attendance tracking with eligibility warnings

Enrollments & Medical Submissions

All the enrollments and medical submissions are digitalized

AI Assistant

Personalized guidance, predictions, and academic support

02 Background

Current challenges faced by students at the Faculty of Engineering that necessitate a comprehensive performance evaluation system.

Attendance Unawareness

Students often don't realize they've fallen below the 80% attendance requirement

⚠️ Course non-eligibility

Course & Minor Confusion

No clear info about modules, electives, or minor-specific subjects

⚠️ Wrong subject selection

Scattered Information

Academic records spread across multiple systems

⚠️ Difficulty tracking progress

Lack of Proactive Guidance

No system exists to warn students or predict outcomes

⚠️ Missed interventions

03

Aim & Objectives

Project Aim

→ The goal is to give students full control and clear visibility over their academic journey using one platform.

Specific Objectives

Academic Tracking

- ✓ Display course history with CGPA
- ✓ Semester-wise performance
- ✓ Track credits

Attendance Management

- ✓ Real-time monitoring
- ✓ Eligibility calculations
- ✓ Proactive warnings

AI Assistant Features

- ✓ Answer academic queries
- ✓ Predict results
- ✓ Motivational suggestions

Additional Features

- ✓ Feedback system
- ✓ Course enrollment, Exam Registrations
- ✓ Time tables

04

Datasets Overview

Required data for system functionality



Student Information

student_id, reg_num, name, batch,
department

PURPOSE: Identify student & link records

HOW TO GET: Request official permission
from Faculty



Course Results

student_id, course_id, marks

PURPOSE: Calculate GPA/CGPA

HOW TO GET: Faculty and Department
offices



Attendance Records

course_id, student_id, lecture_date

PURPOSE: Track attendance percentages

HOW TO GET: Attendance sheets



Course Information

Course_code, name, credits, minor
categories, semester

PURPOSE: Guide course & minor selections

HOW TO GET: Faculty handbook, Faculty
offices



Feedback Data

Student feedback on courses

PURPOSE: Enhance AI, Get student feedbacks

HOW TO GET: Surveys, Feedback feature



Lecturer Information

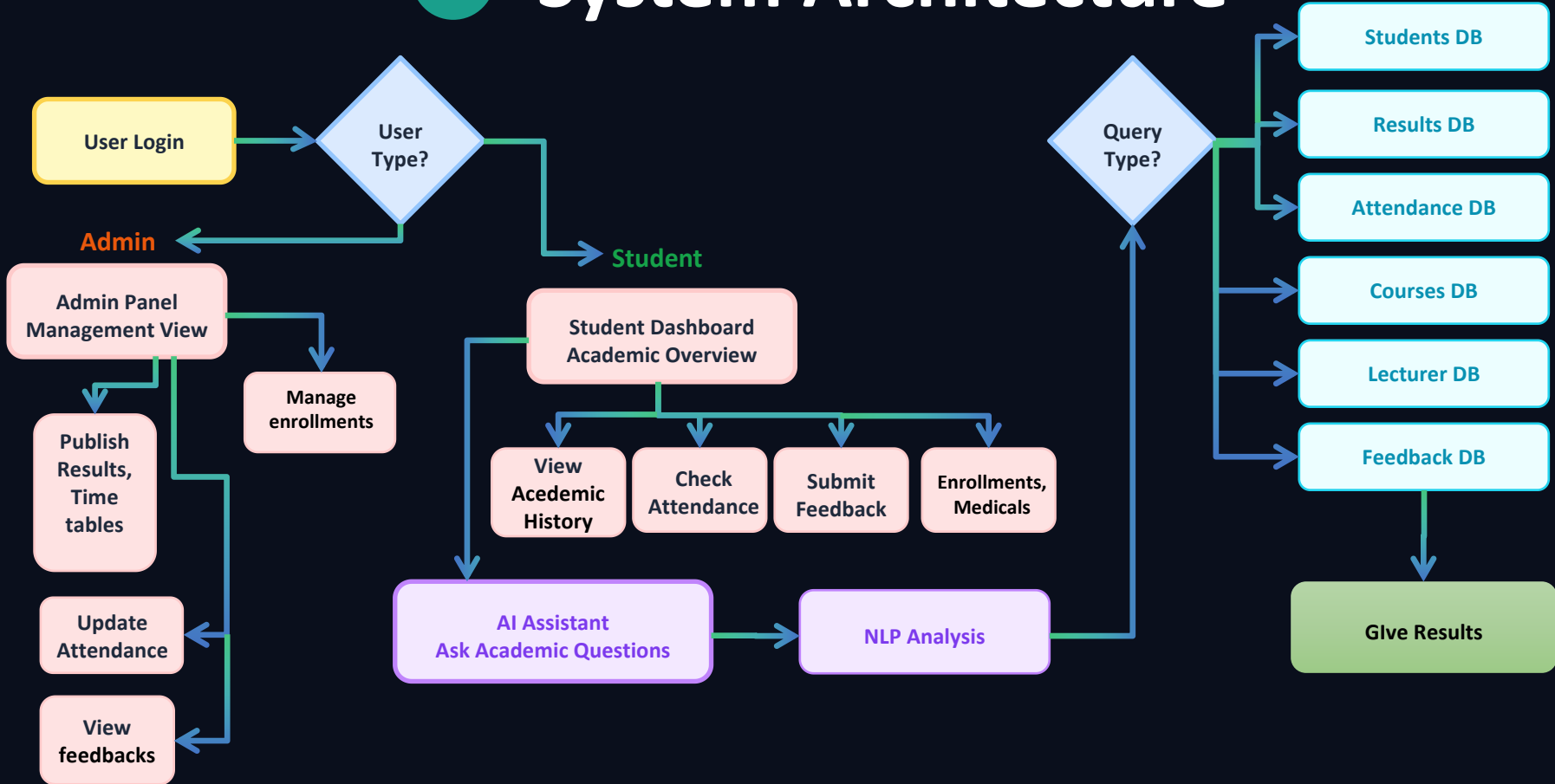
Lecturer ID, name, department, email

PURPOSE: Identify lecturer records

HOW TO GET: Faculty handbook, Faculty offices

05

System Architecture



06

Implementation Plan

Technology Stack

Frontend

React / Next.js

Backend

Node.js / Express

Database

PostgreSQL, NoSQL

AI/ML

Python / Decision
Tree (CART)

Cloud

AWS / Azure

Development Phases

Phase 1

Foundation

- Database design
- User auth
- Basic UI

Phase 2

Core Features

- Academic records
- Attendance
- Enrollment

Phase 3

AI Integration

- AI assistant
- Predictions
- Warnings

Phase 4

Enhancement

- Feedback
- Analytics
- Optimization

07 Milestones & Timeline

A 12-month development plan with clear milestones and deliverables.

Month 1-2

Research & Planning

Requirements, Literature review, Design

Month 3-4

Database & Auth

Schema design, Auth, API development

Month 5-6

Core Modules

Records, Attendance, Enrollment

Month 7-10

AI Integration

Assistant, Predictions, Warnings

Month 11-12

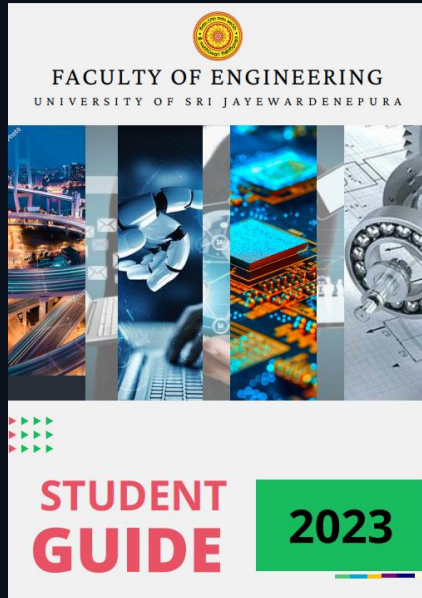
Testing & Deployment

User testing, Bug fixes, Optimization, Deploy

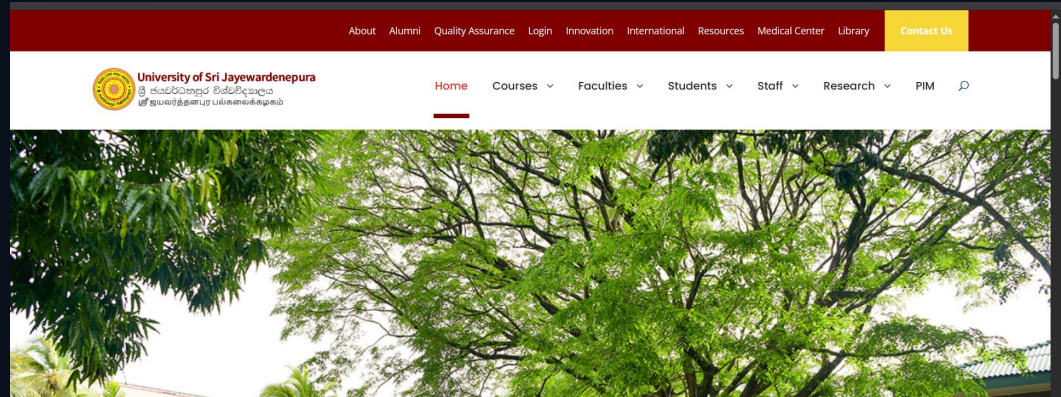
08

References

Faculty Handbook.



Faculty website: <https://www.sjp.ac.lk/>



THANK YOU